

The leading constant in the perimeter deficit of randomized integer convex hulls

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For a smooth planar convex body K of positive curvature and a uniformly rotated and translated integer lattice, the expected perimeter deficit of the integer hull of RK appears to be $C_\infty R^{(-1/3)} \cdot \int \kappa^{(4/3)} ds + o(R^{(-1/3)})$. A heuristic derivation gives $C_\infty = (6/\pi^2) \cdot \int_0^\infty a \cdot G(a) da \approx 0.7190$ for an explicit local function G . Hull data give 0.7198 ± 0.0014 , and the same framework reproduces the published vertex-count constant of Balog and Deshouillers to $3 \cdot 10^{-5}$. A second-order term in $R^{(-1/2)} \cdot \int \kappa^{(3/2)} ds$ is observed but not explained. This is a research note, not a proof.

1. Setting and conjecture

The conjecture is a two-term law for the mean perimeter deficit, local in the curvature of the boundary.

Let $K \subset \mathbb{R}^2$ be a compact convex body with C^3 boundary and curvature $\kappa > 0$, and write $I_\alpha(K) = \int \text{over } \partial K \text{ of } \kappa^\alpha ds$. Thus I_0 is the perimeter, and $I_\alpha = 2\pi$ for the unit disc. Let $L = \rho(\theta)(\mathbb{Z}^2 + t)$, with $\rho(\theta)$ the rotation by θ uniform on $[0, 2\pi)$ and t uniform on $[0, 1)^2$. Since $\rho(\pi/2)\mathbb{Z}^2 = \mathbb{Z}^2$, $\theta \in [0, \pi/2)$ gives the same law. Let $Q_R = \text{conv}(L \cap RK)$ and $\Delta(R) = P(RK) - P(Q_R)$.

Conjecture. As $R \rightarrow \infty$,

$$\mathbb{E}\{\Delta(R)\} = C_\infty R^{-1/3} I_{4/3}(K) + B R^{-1/2} I_{3/2}(K) + o\left(R^{-1/2}\right)$$

with C_∞ and B absolute constants. The data give $C_\infty \approx 0.719\text{--}0.720$ and $B \approx 0.29\text{--}0.30$; §3–§4 predict $C_\infty \approx 0.7190$. The second exponent is tentative (§6).

Equivalently, $C_{\text{eff}}(R) = \mathbb{E}\Delta(R) \cdot R^{(1/3)} / I_{4/3}(K) = C_\infty - B \cdot (I_{3/2} / I_{4/3}) \cdot R^{(-1/6)} + \dots$. The leading term is local: the curvature of RK is κ/R , so $\int \partial(RK) \kappa^{(4/3)} ds = R^{(-1/3)} \cdot I_{4/3}(K)$.

Ngoc and Reitzner [NR] study the expected mean width of this randomized integer hull (in the plane, $P = \pi \cdot W$) and its asymptotic order. A literature search on 6 October 2026 found no value of C_∞ : only the order $R^{(-1/3)}$ is established [NR, BM], while explicit constants are known for the number of vertices [BD, H] (§5). The law was first fitted to simulations [Z1]: the disc, ellipses with axis ratios 2:1 and 5:1 and the superellipse $|x|^3 + |y|^3 = 1$, for R from 1 230 to 10^6 , in 8 448 hulls.

2. The local model and the function G

$G(a)$ is the mean contribution of one hull edge in a rescaled parabolic model; it has a closed form for $0 < a \leq 2$.

Model. Fix $a > 0$. Let δ, s, β be independent and uniform on $[0, 1/a)$, $[0, a)$ and $[0, a)$. Row $k \geq 0$ lies at depth $d_k = \delta + k/a$ and carries the points $s + k\beta + a\mathbb{Z}$. Its admissible part is the set of those points in the closed window $[-v(2d_k), v(2d_k)]$. Let k^* be the first row with a non-empty admissible part, d its depth and $x_1 \leq x_2$ its extreme points. Then

$$c = d \int_{x_1}^{x_2} \left(\frac{x_2^3 - x_1^3}{3} \right) dx, \quad G(a) = \mathbb{E}[c].$$

Meaning. The rows are the lines of the unimodular affine lattice $\Lambda = \{(s + aj + k\beta, \delta + k/a) : j, k \in \mathbb{Z}\}$ parallel to its primitive vector $(a, 0)$. The region is the parabola $\{Y \geq X^2/2\}$. The lowest admissible row is the edge of $\text{conv}(\Lambda \cap \{Y \geq X^2/2\})$ parallel to $(a, 0)$. Since $x_1 \leq 0 \leq x_2$ whenever the row has two points or more,

$$c = \int_{x_1}^{x_2} \min_{p \in \Lambda \cap [x_1, x_2]} \left(d - \frac{p^2}{2} \right) dp, \quad d \geq \frac{a}{2},$$

the contribution of that edge to Cauchy's perimeter integral (§3). A single point gives $c = 0$.

Closed form for $0 < a \leq 2$. This is derived in [Z2] for $a \in [1, 2]$; the derivation holds unchanged on $(0, 2]$.

- $k^* \in \{0, 1\}$, because row 1 always meets its window when $a^3 \leq 8$.
- Write $c = f(x_2) - f(x_1)$ with $f(x) = dx - x^3/3$, which is odd. For a uniform phase and $r = v(2d) \geq a/2$, x_2 is uniform on $[r - a, r]$, so the phase average is $H(d, a) = (2r - a)(r^2 + 2ar - a^2)/6$; $H = 0$ when $r < a/2$.
- With $h = a^2/8$, $G = G_0 + G_1$, where

$$\begin{aligned} G_0(a) &= a \int_{h^{1/a}}^{\infty} H(d, a) dd = \frac{4\sqrt{2}}{15} a^{-3/2} + \frac{12}{9} a^{3/2} + \frac{a^3}{6} - \frac{17a^6}{5760}, \\ G_1(a) &= a \int_0^h \left(1 - \frac{2\sqrt{2\delta}}{a} \right) H\left(\frac{\delta}{a}, \frac{a}{\delta}\right) d\delta. \end{aligned}$$

The closed form agrees with a direct Monte Carlo simulation of the model ($2 \cdot 10^6$ draws per value, points enumerated row by row):

a	$G(a)$, closed form	$G(a)$, Monte Carlo
0.3	2.703741	2.703779 ± 0.001402

a	G(a), closed form	G(a), Monte Carlo
0.7	0.863439	0.863427 ± 0.000467
1.5	0.145227	0.145461 ± 0.000186
2.0	0.034677	0.034681 ± 0.000117

For $a > 2$ the first admissible row can lie deep (k^* reaches about $a^3/8$), and G is computed by simulation (§4).

3. Heuristic derivation of C_∞

Five steps, two of them exact, give $C_\infty = (6/\pi^2) \cdot J$ with $J = \int_0^\infty a \cdot G(a) da$.

1. **Cauchy's formula (exact).** With h the support function and φ the outer normal angle,

$$P(RK) - P(Q_R) = \int_0^{2\pi} \big(h_{\{RK\}} - h_{\{Q_R\}}\big)(\varphi) d\varphi.$$

2. **Local rescaling (heuristic: parabolic approximation).** Near the boundary point of normal φ and curvature κ , set $x = \kappa^{-1/3} \cdot X$ along the tangent and $y = \kappa^{1/3} \cdot Y$ along the inner normal. The boundary becomes $Y = X^2/2$ to leading order, and the lattice becomes a unimodular affine lattice. Support deficits scale by $\kappa^{1/3}$, and $d\varphi = \kappa^{2/3} \cdot d\theta$ with θ the rescaled slope.

3. **Edge decomposition (exact in the parabolic model).** At slope θ , a hull vertex p supports the deficit $D_p(\theta) = Y_p - \theta X_p + \theta^2/2$. This expression is invariant under the shears $(X, Y) \mapsto (X - t, Y - tX + t^2/2)$, which preserve the parabola and unimodularity.

- Give each edge $e = (p_1, p_2)$ the slope interval $[X(p_1), X(p_2)]$. These intervals tile the line.
- In the frame where e is horizontal, the integral of $\min(D(p_1), D(p_2))$ over that interval is the c of §2.
- The minimum switches exactly at the slope of e . So each vertex receives exactly the slopes between its two adjacent edges, and the edge sum equals Cauchy's integral.

4. **Counting edges (heuristic: equidistribution).** The edge at normal φ is parallel to a primitive vector v of L .

- Per unit normal angle, primitive vectors with $|v| \in [\ell, \ell + d\ell]$ have density $(6/\pi^2) \cdot \ell d\ell$ (Siegel's mean value theorem [S]; asymptotically exact for the rotated $\mathbb{Z}^2$).
- In the rescaled frame, v has length $a = \kappa^{1/3} \cdot |v|$.
- The offsets (s, δ) are exactly uniform, because t is.

- The shear β is set by the Farey neighbour of v . We assume it is asymptotically uniform on $[0, a)$ over vectors of comparable length. Then $E[c | a] = G(a)$.

5. Assembly. Per unit normal angle, the expected edge sum is $\kappa \cdot (6/\pi^2) \cdot \int \ell \cdot G(\kappa^{1/3} \ell) d\ell = (6/\pi^2) \cdot \kappa^{1/3} \cdot J$. Integrating against $d\phi = \kappa ds$ gives

$$\mathbb{E}[\Delta(R)] \simeq \frac{6}{\pi^2} J \int_{\partial(R)} \kappa^{4/3} ds = \frac{6}{\pi^2} J I_{4/3}(K), R^{-1/3},$$

$$C_{\infty} = \frac{6}{\pi^2} \int_0^{\infty} a G(a) da.$$

Equivalent form. Applying Cauchy's formula direction by direction, C^∞ should also be the mean depth of the lowest point of a Haar-random unimodular affine lattice inside $\{Y \geq X^2/2\}$. Step 3 and Siegel's theorem should identify the two forms; this was not checked numerically here.

Not proved. Three points remain open.

- Equidistribution of the rotated and stretched lattices, i.e. expanding circles in the space of affine lattices [EM, MS], for test functions that are unbounded near the cusp.
- Uniform validity of the parabolic approximation.
- The limit $R \rightarrow \infty$ in the presence of the heavy tail $a \cdot G(a) \sim 1/(9a^2)$ (§4), which comes precisely from the cusp.

4. Evaluation of J and C^∞

J lies between 1.1823 and 1.1831, so $C^\infty = (6/\pi^2) \cdot J$ lies between 0.7188 and 0.7193. The closed form on $(0, 2]$ supplies 94 % of J.

Range of a	Method	Contribution to J
(0, 1]	Closed form, quadrature to 25 digits	0.87247264
[1, 2]	Closed form; rigorous enclosure 0.23468926600629... ($\pm 2 \cdot 10^{-49}$)	0.23468927
[2, 3.6]	Monte Carlo, 17 values of $a \times 4 \cdot 10^6$ draws, Simpson rule	0.043081 ± 0.000062
[3.6, ∞)	Importance sampling up to $a \approx 20$, then the asymptote below	0.0321 to 0.0329
Total J		1.1823 to 1.1831
$C^\infty = (6/\pi^2) \cdot J$		0.7188 to 0.7193

The [1, 2] piece is the integral certified in [Z2] to 6 decimals; ball arithmetic (Arb) reproduces it to 41 digits in under a second.

Tail. For large a , $c \neq 0$ requires a row with two admissible points before any row with one. With row phases $s + k\beta$, this forces $\beta \equiv 0 \pmod{a}$ to within $O(a^{-2})$. Near the last empty row, the admissible pairs (phase, β) form a funnel tip of area $16f(1 - f)/a^4$, where $f \in [0, 1)$ is the fractional row position of the depth $a^2/8$. On that set $c \approx a^3/24$, and $E[f(1 - f)] = 1/6$, so

$$a \backslash, G(a) \sim \frac{1}{9} \backslash, a^{\{2\}} \quad (a \rightarrow \infty).$$

Simulation agrees: $a^2 \cdot (a \cdot G)$ is 0.118 at $a \approx 5.9$, 0.116 at $a \approx 7.5$, 0.10 ± 0.01 at $a \approx 11$ and 0.09 ± 0.03 at $a \approx 16$. Two tail fits bracket the last row of the table: one imposes the asymptote $1/9$ with corrections in $1/a$ and $1/a^2$, the other leaves the amplitude free. Their spread is the main uncertainty on C^∞ .

5. Comparison with hull data

The prediction 0.7188–0.7193 agrees with the three-parameter fit of the positive-curvature data, $C^\infty = 0.7198 \pm 0.0014$.

The data are the means of [Z1] for the disc and the ellipses 2:1 and 5:1, each at 23 radii from 1 230 to 10^6 (69 points). The fitted model is $C_{\text{eff}}(R) = C^\infty - B \cdot (I_3/2/I_4/3) \cdot R^{(-\gamma)}$, with $I_3/2/I_4/3 = 1, 1.128$ and 1.488 for the three shapes.

Fit	C^∞	γ	B	χ^2 per d.o.f.
69 points, all free	0.7198 ± 0.0014	0.169 ± 0.005	0.300 ± 0.009	1.25
69 points, C^∞ fixed at 0.7193	fixed	0.171 ± 0.002	0.302 ± 0.007	1.23
Disc only, all free	0.7137 ± 0.0037	0.199 ± 0.017	0.356 ± 0.033	1.24
Disc only, C^∞ fixed at 0.7193	fixed	0.176 ± 0.003	0.319 ± 0.009	1.27
92 points with $p = 3$, $\gamma = 1/6$ fixed [Z1]	0.7204 ± 0.0005	fixed	0.296 ± 0.003	1.13

Errors are nominal, from a diagonal covariance: the covariance between radii cannot be estimated from the published means. The cut $R \geq 1\,230$ and the model form were not preregistered. C^∞ and B are correlated (-0.65 in the 69-point fit, -0.94 for the disc alone).

An independent simulation, with its own hull code, reproduces the published means for the disc:

R	This note	[Z1]	Difference
1 230	0.6303 ± 0.0011 (400 hulls)	0.6293 ± 0.0025	$+0.4 \sigma$
9 587	0.6558 ± 0.0009 (300 hulls)	0.6581 ± 0.0021	-1.0σ
110 000	0.6779 ± 0.0008 (120 hulls)	0.6794 ± 0.0007	-1.3σ

Check against a published constant. The same framework predicts the mean number of vertices. Replacing c by the indicator that the lowest admissible row holds at least two points, of probability $p(a)$, gives $E V \approx (6/\pi^2) \cdot J_V \cdot \int \kappa^{1/3} ds$ with $J_V = \int_0^\infty a \cdot p(a) da$. For the disc of radius R this is $(12/\pi) \cdot J_V \cdot R^{2/3}$. The closed form on $(0, 2]$, simulation on $[2, 4]$ and the tail $a \cdot p(a) \sim 8/(3a^5)$ give $(12/\pi) \cdot J_V = 3.45360 \pm 0.00014$.

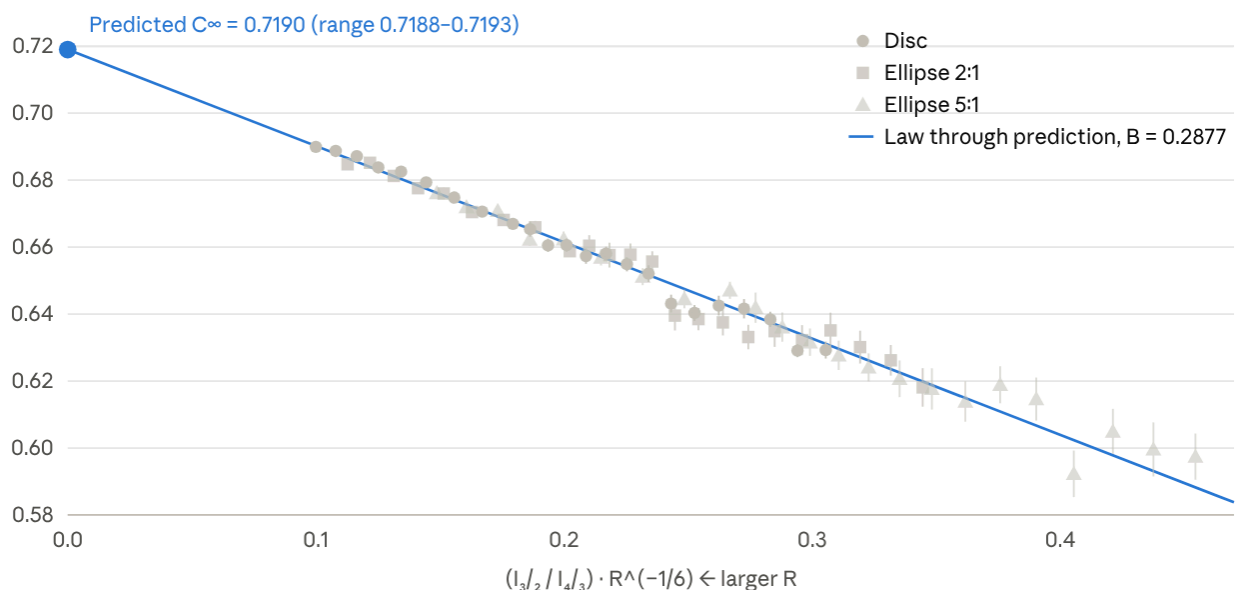
Balog and Deshouillers proved that the number of vertices of $\text{conv}(\mathbb{Z}^2 \cap rD)$, averaged over the radius, is asymptotic to $3.4536898915 \dots \cdot R^{2/3}$ [BD, F]. The two values agree to $3 \cdot 10^{-5}$. This supports the counting step of §3 (density $6/\pi^2$, uniform shear), which the perimeter prediction shares.

Extrapolation to $R = \infty$

Hull data extrapolate to the predicted constant $C^\infty = 0.7190$

Mean constant C_{eff} from [Z1], ± 1 s.e., against $(I_3/I_2 / I_4/I_3) \cdot R^{1/6}$; $x = 0$ is $R = \infty$.

Blue: the predicted C^∞ of §4 and the two-term law through it, with B fitted and exponent $1/6$.



Disc: $R = 1\,230$ at $x \approx 0.31$ and $R = 10^6$ at $x = 0.10$; the ellipses sit further right by their curvature ratio.

[Z1] reanalysis means, 69 points, $R = 1\,230$ to 10^6 · prediction from §4 · line: C^∞ fixed, exponent $1/6$, B fitted (χ^2 per d.o.f. 1.30)

In these coordinates the two-term law is a straight line. As R grows, the means of all three shapes move left along one line toward the axis, where it meets the predicted constant.

6. The second-order term (open)

The data call for a correction $-B \cdot R^{(-1/2)} \cdot \int \kappa^{(3/2)} ds$ with $B \approx 0.30$. A lattice-cutoff mechanism produces exactly this structure, but only about a fifth of B .

Data. With C^∞ fixed at 0.7193, the 69 points give an exponent $\gamma = 0.171 \pm 0.002$ in C_{eff} , close to $1/6$, and $B = 0.302 \pm 0.007$ (§5).

Mechanism. The tail $a \cdot G \sim 1/(9a^2)$ comes from shears $\beta \equiv 0 \pmod{a}$ within $O(a^{-2})$, that is, from a very short Farey neighbour w of v . In $\mathbb{Z}^2$, $|w| \geq 1$, which removes shears $|\beta \bmod a|$ below $\varepsilon = \kappa^{(1/3)}$ in the rescaled frame. Write b for the shear β reduced to $(-a/2, a/2]$; summed over a , the contribution density in b behaves as $c_0 \cdot |b|^{(-1/2)}$ near $b = 0$. Removing a band of width ε therefore costs a relative $\varepsilon^{(1/2)} = \kappa^{(1/6)}$, which is exactly the structure $R^{(-1/2)} \cdot \int \kappa^{(3/2)} ds$.

Size. The funnel analysis of §4 gives $c_0 = \sqrt{2}/30$ (numerically, to 20 digits). Short neighbours follow the discrete length distribution of the primitive vectors of $\mathbb{Z}^2$. Zeta regularization of the resulting sum over primitive w gives

$$\begin{aligned} B_{\mathrm{cut}} &= -\frac{c_0}{\pi} \backslash, \\ Z_{\mathrm{prim}} &\backslash \left(\frac{3}{4} \right), \\ &\quad \\ Z_{\mathrm{prim}}(s) &= \sum_{w \in \mathrm{primitive}} |w|^{-2s} = \\ &\frac{4}{\pi} \backslash, \zeta(s) \backslash, \beta(s) \backslash \zeta(2s), \end{aligned}$$

where $\beta(s)$ is Dirichlet's beta function. Numerically, $Z_{\text{prim}}(3/4) = -3.8576$ and $B_{\text{cut}} = 0.0579$. The regularized constant was checked by direct summation over $|w| \leq 3\,200$: -1.00982 against -1.00992 predicted for $(\pi/12) \cdot Z_{\text{prim}}(3/4)$.

Open. The cutoff explains about a fifth of B , so other corrections of comparable size must be present. A plausible but unchecked candidate is the error term in the equidistribution of Farey neighbours. Hyperbolic lattice-point counting would suggest a relative order near $R^{(-2/9)}$, which is hard to separate from $R^{(-1/6)}$ over the simulated range.

7. Questions for specialists

Four precise questions, in order of priority:

1. Is the constant C^∞ , or the identity $C^\infty = (6/\pi^2) \cdot \int_0^\infty a \cdot G(a) da$, already known? Our search found only the order of the deficit $[NR, BM]$ and vertex-count constants $[BD, H]$; the full texts of $[BD]$ and $[H]$ were not accessible to us.

2. Can §3 be made rigorous? The key input is equidistribution of expanding circles in $ASL(2, \mathbb{R})/ASL(2, \mathbb{Z})$ for test functions with the cusp growth that produces $a \cdot G(a) \sim 1/(9a^2)$.
3. What is the true second-order term, both its exponent and its constant, and what besides the lattice cutoff of §6 contributes to it?
4. Does the leading formula extend to boundaries with isolated flat points? For the superellipse $|x|^3 + |y|^3 = 1$ the curvature vanishes at four points but $I_4/3$ is finite. Its 23 means in [Z1] fit the two-term law, with the constants of the positive-curvature shapes held fixed, at $\chi^2/n = 0.85$.

8. Data, code, provenance and AI assistance

All numbers in this note can be reproduced from the two deposits and six short scripts, available from the author.

- **Hull data.** [Z1] holds the raw simulations (8 448 randomized hulls, rotation and translation uniform) and a reanalysis of the 92 means.
- **Local model.** [Z2] gives the closed form of G on $[1, 2]$ and the certified value of the $[1, 2]$ piece of J .
- **Scripts.** They cover the evaluation of J (§4), the tail by importance sampling, the second-order estimate (§6), the fits, the independent disc simulation and the vertex-count check (§5). They need Python with NumPy, SciPy, mpmath and pandas, and run in about fifteen minutes in total.
- **AI assistance.** The simulations and the local model of [Z1, Z2] were developed by the author with substantial help from ChatGPT/Codex. The derivation of §3, the tail asymptotics, the evaluation of J , the analysis of §6 and the independent simulations were produced with Claude (Anthropic). The author is responsible for the content.
- **Review status.** No independent human mathematical review has yet taken place.

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[NR], [B], [BM], [F] and the abstract of [H] were consulted online on 6 October 2026. The other references are cited from standard bibliographic knowledge, and their details should be checked before circulation.